

Power-sum lower bounds on hidden eigenvalues in positive realizations

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Abstract

A positive realization of a transfer function of order n is a nonnegative matrix of some size $N \geq n$ carrying the n poles among its eigenvalues; the $s = N - n$ remaining eigenvalues are the *hidden* eigenvalues, whose characterization Benvenuti and Farina list as an open problem. We give a lower bound on s from the power sums of the poles: if p_k denotes the k -th power sum and ρ the spectral radius, then $s \geq \max_{k \geq 1} \lceil [-p_k]_+ / \rho^k \rceil$. At $k = 1$ this reproduces the quantity ζ of Benvenuti's lower bound; for $k \geq 2$ it is, as far as we can determine, new, and it is strictly stronger. On the family $\mathcal{M}_m(\theta) = \{1\} \cup m \cdot \{\pm i\theta\}$ the published lower bound is vacuous ($N \geq n$) while ours gives $N \geq 2m(1 + \theta^2)$; when $\theta^2 > 1 - 1/2m$ this meets Benvenuti's own upper bound $4m$ and therefore determines the minimal size exactly, where previously only $n \leq N \leq 4m$ was known. The bound is attained: on $\{1\} \cup m \cdot \{\pm i\}$ it gives $N \geq 4m$, and a direct sum of m copies of the 4-cycle permutation matrix realizes $N = 4m$. We also record two limitations. The method has a hard ceiling — it can never prove more than $N \leq 2n$ — so it is complementary to, not competitive with, the digraph bounds that give unbounded ratios; and the ratio 2 itself is not new territory, being exactly what the Benvenuti–Farina family of 1999 achieves. Finally we explain why no such bound can exist in the Boyle–Handelman/Kim–Ormes–Roush setting: there one appends only zeros, and zeros change no power sum. All claims are machine-verified in exact rational arithmetic.

1 Introduction

Let $H(z)$ be a rational transfer function of order n with pole multiset $\Lambda = \{\lambda_1, \dots, \lambda_n\}$ and spectral radius $\rho = \max_i |\lambda_i|$. A *positive realization* of H is a triple (A, b, c) with $A \in \mathbb{R}_{\geq 0}^{N \times N}$, $b, c \geq 0$ and $H(z) = c(zI - A)^{-1}b$. Existence is settled [8, 1, 13], and a clean criterion is available in terms of an invariant polyhedral cone [5]. What is not settled is *size*. The poles of H are a subset of the eigenvalues of any positive realization, and the remaining $s = N - n$ eigenvalues — the *hidden eigenvalues* — are the reason N may exceed n . Benvenuti and Farina put it as follows [5, §IX]: “An important issue related to minimality of positive systems is the study of ‘hidden’ eigenvalues, i.e. of the eigenvalues which are not poles and that possibly one has to add in order to obtain a minimal positive realization. A full characterization of this property may lead to a deeper and valuable insight into the problem.” The minimality problem remains open; a recent statement of its status is [14].

This note contributes one inequality. It is elementary, and its only merit is that it appears not to have been written down beyond its first instance.

The bound. Write $p_k = p_k(\Lambda) = \sum_i \lambda_i^k$ for the k -th power sum of the poles and $[x]_+ = \max(x, 0)$. Theorem 1 states

$$s \geq \left\lceil \max_{k \geq 1} \frac{[-p_k]_+}{\rho^k} \right\rceil. \quad (1)$$

What is already known. The case $k = 1$ of (1) is published. Benvenuti [2, Thm. 4.7] proves $N \geq N_D + \sum_{|s| < 1} m(s) + \zeta$ with $\zeta = \lceil -m_{\mathcal{R}_1} - \sum_{|s| < 1} s m(s) \rceil$ when that quantity is positive, and the proof is exactly the $k = 1$ argument: “since the trace of a nonnegative matrix is nonnegative ... since each one of these additional eigenvalues has to be not greater than 1, then their number is not lesser than ζ ” [2, pp. 379–380]. The companion paper [3] transfers the same quantity to the realization dimension. We have found no power sum beyond the first anywhere in this literature: not in [2, 3], not in the digraph bound of Nagy and Matolcsi [12], not in [5, 14]. Section 3 shows that $k = 2$ already does strictly better.

What we do not claim. Two disclaimers, stated here rather than buried.

- (a) The ratio $N/n = 2$ is *not* new territory. The family $H(z, m) = 1/((z - 1)(z^{2^m} + 1))$ of Benvenuti and Farina [4] has $n = 2^m + 1$ and minimal positive realization of dimension $N = 2^{m+1} = 2n - 2$, so ratio 2 is exactly what they already achieve; and Nagy and Matolcsi [12] obtain *unbounded* ratios, with $n = 4$ fixed and $N \geq m + 1$ for arbitrary m .
- (b) Worse, ratio 2 is a hard ceiling for the present method. Since $p_k \geq -\sum_i |\lambda_i|^k \geq -n\rho^k$, the right-hand side of (1) never exceeds n , so (1) can never prove more than $N \leq 2n$ (Proposition 3). The bound is therefore complementary to the combinatorial bounds of [12], not a competitor to them.

What we do claim is (i) the inequality for $k \geq 2$, (ii) that it strictly improves the published lower bound, and (iii) that on an explicit family it closes the published gap completely.

2 The bound

Theorem 1. Let $A \in \mathbb{R}_{\geq 0}^{N \times N}$ and let Λ be a sub-multiset of $\text{spec } A$ of size n containing an eigenvalue of modulus $\rho(A)$; write $\rho = \rho(A)$ and $s = N - n$. Then for every $k \geq 1$

$$s \geq \frac{-p_k(\Lambda)}{\rho^k}, \quad \text{hence} \quad s \geq \left\lceil \max_{k \geq 1} \frac{[-p_k(\Lambda)]_+}{\rho^k} \right\rceil.$$

Proof. Let $\mathcal{M}$ be the complementary multiset, $|\mathcal{M}| = s$, so that $\text{spec } A = \Lambda \uplus \mathcal{M}$. Every eigenvalue of A has modulus at most $\rho(A) = \rho$, so $|\sum_{\mu \in \mathcal{M}} \mu^k| \leq \sum_{\mu \in \mathcal{M}} |\mu|^k \leq s\rho^k$. Since A is nonnegative, $\text{Tr}(A^k) \geq 0$: it is a sum of products of nonnegative entries. Therefore

$$0 \leq \text{Tr}(A^k) = p_k(\Lambda) + \sum_{\mu \in \mathcal{M}} \mu^k \leq p_k(\Lambda) + s\rho^k,$$

which rearranges to the claim; s is an integer, so the ceiling may be taken. $\square$

The hypothesis that Λ contains a dominant eigenvalue is what makes ρ computable from the poles alone; in the realization setting it holds because $\rho(H)$ is a pole [5, Lem. 7].

Proposition 2 ($k = 1$ is Benvenuti's ζ). *At $k = 1$, (1) reads $s \geq \lceil -p_1(\Lambda)/\rho \rceil$, which after normalising $\rho = 1$ and splitting Λ into its dominant and non-dominant parts is the quantity ζ of [2, Thm. 4.7].*

Proof. Normalise $\rho = 1$. The dominant part of Λ contributes $m_{\mathcal{R}_1}$ to p_1 , since the h -th roots of unity sum to zero for $h > 1$ [2, proof of Thm. 4.7], and the non-dominant part contributes $\sum_{|s|<1} s m(s)$. Hence $-p_1 = -m_{\mathcal{R}_1} - \sum_{|s|<1} s m(s)$, and the ceiling of its positive part is ζ . $\square$

Proposition 3 (Ceiling of the method). *The right-hand side of (1) is at most n ; consequently Theorem 1 can never certify $N > 2n$.*

Proof. $-p_k \leq \sum_i |\lambda_i|^k \leq n\rho^k$. $\square$

3 Strict improvement, and an exact determination

Fix $0 < \theta < 1$ and $m \geq 1$, and let

$$\mathcal{M}_m(\theta) = \{1\} \cup m \text{ copies of } \{i\theta, -i\theta\}, \quad n = 2m + 1, \quad \rho = 1.$$

Only 1 is dominant, so in the notation of [2] we have $r = 1$, $N_D = 1$ and $m_{\mathcal{R}_1} = 1$.

Theorem 4. *For $\mathcal{M}_m(\theta)$:*

(i) *the lower bound of [2, Thm. 4.7] gives $N \geq 2m + 1 = n$, i.e. it is vacuous, because $\zeta = 0$;*

(ii) *Theorem 1 at $k = 2$ gives $s \geq \lceil 2m\theta^2 - 1 \rceil$, hence $N \geq 2m + 1 + \lceil 2m\theta^2 - 1 \rceil$, which exceeds n whenever $\theta^2 > 1/2m$;*

(iii) *the upper bound of [2, Thm. 4.4] gives $N \leq 4m$;*

(iv) *if $\theta^2 > 1 - \frac{1}{2m}$ then the bounds in (ii) and (iii) coincide and*

$$N_{\min} = 4m.$$

Proof. (i) $p_1(\mathcal{M}_m(\theta)) = 1$ and the non-dominant numbers cancel in conjugate pairs, so $m_{\mathcal{R}_1} + \sum_{|s|<1} s m(s) = 1 \geq 0$ and $\zeta = 0$; the bound reads $N \geq N_D + 2m + 0 = 2m + 1$. (ii) $p_2 = 1 + m((i\theta)^2 + (-i\theta)^2) = 1 - 2m\theta^2$, so $[-p_2]_+/\rho^2 = 2m\theta^2 - 1$ when positive. (iii) is [2, Thm. 4.4] with $\kappa(i\theta) = 4$, valid since $i\theta \in \Pi_4 \setminus \Pi_3$ for $\theta > 1/\sqrt{3}$. (iv) The bound in (ii) equals $4m$ as soon as $\lceil 2m\theta^2 - 1 \rceil \geq 2m - 1$, i.e. $2m\theta^2 - 1 > 2m - 2$, i.e. $\theta^2 > 1 - 1/2m$. $\square$

Table 1 shows the three quantities. The criterion $\theta^2 > 1 - 1/2m$ of Theorem 4(iv) predicts exactly which rows close: it holds for $\theta = 9/10$ only at $m = 1$ ($0.81 > 0.5$) and fails at $m = 4$ ($0.81 < 0.875$), where the bound indeed falls one short of the upper bound; it holds throughout for $\theta = 99/100$ from $m = 2$ on and for $\theta = 999/1000$, and there the gap closes and N is determined.

θ	m	n	[2, Thm. 4.7] lower	Theorem 1 lower	[2, Thm. 4.4] upper
9/10	1	3	3	4	4
9/10	4	9	9	15	16
9/10	16	33	33	58	64
99/100	2	5	5	8	8
99/100	4	9	9	16	16
99/100	8	17	17	32	32
99/100	16	33	33	64	64
999/1000	4	9	9	16	16
999/1000	16	33	33	64	64

Table 1: Lower and upper bounds on the minimal size N for $\mathcal{M}_m(\theta)$. Bold: the new lower bound meets the published upper bound, so N is determined exactly. All entries exact.

4 Sharpness

Let $\mathcal{M}_m = \{1\} \cup m \cdot \{i, -i\}$, all of modulus one, $n = 2m + 1$.

Theorem 5. $p_2(\mathcal{M}_m) = 1 - 2m$, so Theorem 1 gives $s \geq 2m - 1$ and $N \geq 4m$. The bound is attained: if C_4 is the 4-cycle permutation matrix, with characteristic polynomial $x^4 - 1$ and spectrum $\{1, i, -1, -i\}$, then $C_4^{\oplus m}$ is nonnegative of size $4m$ and contains $\mathcal{M}_m$ in its spectrum. Hence $N_{\min} = 4m = 2n - 2$.

Proof. The power sum is immediate. For the construction, $\text{spec } C_4^{\oplus m}$ is m copies of $\{1, i, -1, -i\}$, which contains one 1 and m copies of each of $\pm i$. $\square$

Remark 6. Two caveats. First, for $m \geq 2$ the multiset $\mathcal{M}_m$ violates the standing hypothesis of [2, Thms. 3.2, 4.1] that the spectral radius occur with dominant multiplicity: here $m(1) = 1$ while the dominant multiplicity is m . The formula of [2, Thm. 4.1] nevertheless evaluates to $N_D = 4m$ on $\mathcal{M}_m$, so the conclusion $N_{\min} = 4m$ is also reachable by that route, by an argument through roots of unity rather than through traces; we note this so that Theorem 5 is not mistaken for a new determination. It is a sharpness check on (1), not a new value of N . Second, $4m = 2n - 2$ is the Benvenuti–Farina ratio of [4], as recalled in Section 1.

Remark 7 (Tutorial Example 8). For the poles $\{1, \pm 0.9i\}$ of [5, Ex. 8], $p_2 = -31/50$ and (1) gives $s \geq 1$, i.e. $N \geq 4$. This is attained by the nonnegative circulant with first row $(0, \frac{19}{20}, 0, \frac{1}{20})$, whose characteristic polynomial is $(x - 1)(x + 1)(x^2 + \frac{81}{100})$: the hidden eigenvalue is -1 . The example in [5] is 4-dimensional, so the bound is tight there as well.

5 Why there is no analogue for the Boyle–Handelman problem

It is worth saying why the same traces behave completely differently one category over. In the inverse eigenvalue problem solved by Boyle and Handelman over $\mathbb{R}$ [6] and by Kim, Ormes and Roush over $\mathbb{Z}$ [9], one asks whether Λ is the *nonzero* spectrum of a nonnegative matrix; the matrix may be enlarged, but only by appending *zeros*. Zeros contribute nothing to any power sum, so the trace conditions $\text{tr}_k \geq 0$ must already hold for Λ itself: they are a genuine

obstruction, not a counting device, and no lower bound of the shape (1) on the size of the dilation can exist. Equivalently: in that category the analogue of Theorem 1 is the statement $p_k(\Lambda) \geq 0$, with no s in it.

This is the exact point at which results from symbolic dynamics fail to transfer to the positive realization problem. A negative p_2 closes the fixed-nonzero-spectrum problem outright — shift equivalence preserves the nonzero spectrum with multiplicities [10, §7.4], so no dilation of any size helps, an argument used in this form in [7, II] — while in the present setting the same negative p_2 merely counts hidden eigenvalues, and indeed [5, Ex. 8] exhibits a transfer function with $p_2 = -0.62$ and a 4-dimensional positive realization. The two categories differ exactly by the freedom to add nonzero eigenvalues.

6 Verification

The script `hidden_eigs.py` accompanying this note runs the checks of Table 2 in 23 s, all passing; its output is archived as `hidden_eigs_output.txt`. Arithmetic is exact rational throughout; the only floating-point quantity anywhere is the spectral radius in check (T), and that row is labelled accordingly.

key	check	mode
(T)	Theorem 1 holds on 120 random nonnegative integer matrices, Λ a rational factor	exact + numeric ρ
(K)	at $k = 1$ the bound equals ζ of [2, Thm. 4.7]	exact
(I)	$k = 2$ strictly exceeds [2, Thm. 4.7] in 15/15 instances (Table 1)	exact
(S)	Theorem 5: bound = $4m$ attained by $C_4^{\oplus m}$, $m \leq 32$	exact
(E)	Remark 7: bound = 1, attained by an explicit rational circulant	exact
(C)	Proposition 3: the bound never exceeds n , on 300 random pole sets	exact
(Z)	appending zeros leaves every power sum unchanged (Section 5)	exact

Table 2: The verification battery (`hidden_eigs.py`), 7/7 passing.

In check (T) the spectral radius is replaced by a rational *lower* bound, which makes the tested inequality strictly stronger than the theorem; no violation occurred. The bound was attained in none of those 120 random instances, which is the expected behaviour: like the $k = 1$ bound it is weak on generic spectra and bites only when the poles are strongly oscillatory.

7 Scope

Established here. Theorem 1 for all k ; its coincidence with [2, Thm. 4.7] at $k = 1$; strict improvement and an exact determination of $N_{\min}$ on $\mathcal{M}_m(\theta)$ for $\theta^2 > 1 - 1/2m$; sharpness on $\mathcal{M}_m$ and on [5, Ex. 8]; the ceiling $N \leq 2n$; the structural reason the bound has no analogue in [6, 9].

Open. (a) Whether power sums can be combined with the cone criterion [5] or with the digraph bound [12] to pass the ratio-2 ceiling. (b) Whether the Johnson–Loewy–London inequalities [11], which mix power sums of different orders, give a bound not implied by (1); we have not investigated this and it is the obvious next step. (c) Whether the bound is ever tight outside the families of Sections 3–4; it was not tight in any of our 120 random instances.

Not claimed. Ratio 2, which is [4]; unbounded ratios, which are [12]; and the $k = 1$ case, which is [2].

Provenance

Unrefereed preprint. Theorem 1 is elementary and its proof is complete above. The bounds of [2] quoted for comparison in Table 1 are evaluated by us from the published statements of Theorems 4.4 and 4.7 of that paper and are reproduced in the script; we have not reproved them. The symbolic-dynamics facts in Section 5 are quoted from [10, 6, 9]. We searched the literature for a power-sum bound with $k \geq 2$ and found none; a search cannot establish a negative, and we state this as the limit of our knowledge.

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